

UNIVERSITY OF ENGINEERING AND TECHNOLOGY
VIETNAM NATIONAL UNIVERSITY HANOI

Research on Enhancing Vision-Language Model for Document Key Information Extraction

Weekly report

Presenter

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Major: Information Systems

Supervisor

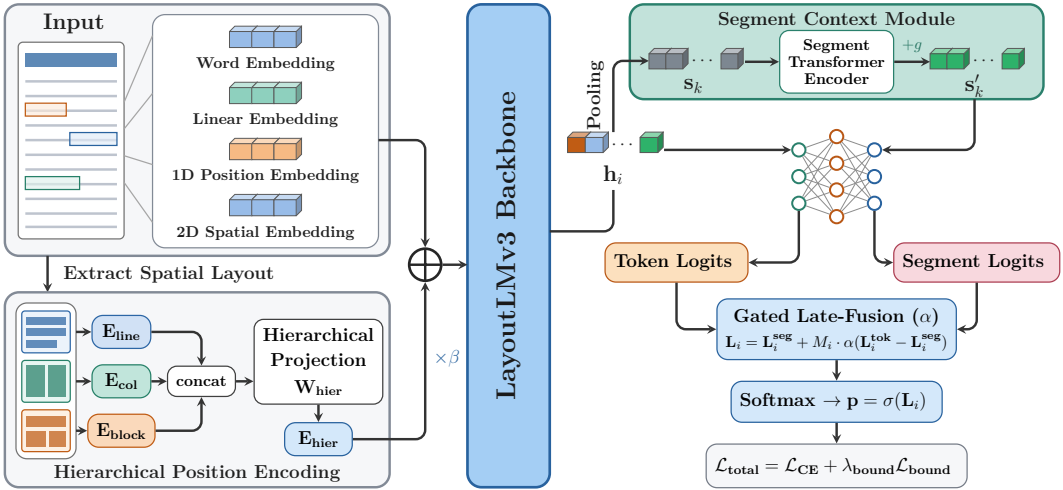
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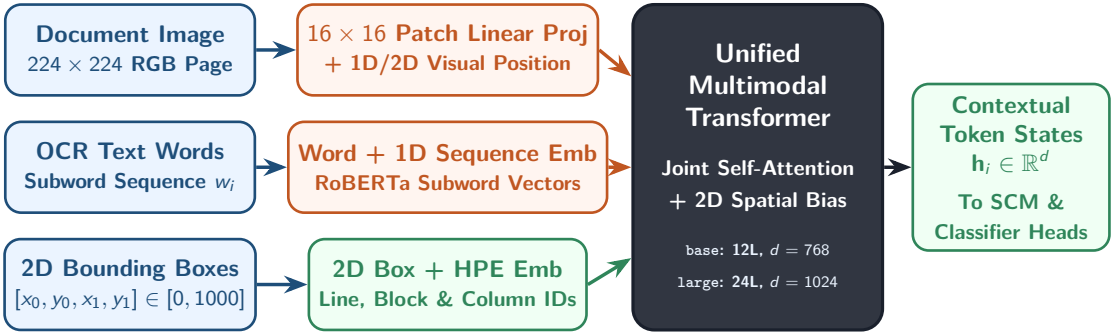
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1. End-to-End Model Architecture



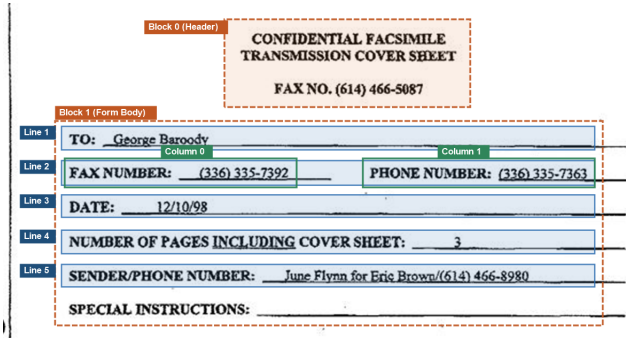
1. LayoutLMv3 Backbone Overview



1. Backbone Limitations & Proposed Solutions

Limitation	Failure Behavior	Real-Life Example	Proposed Solution
1. Flat 2D Coords	Raw $[x_0, y_0, x_1, y_1]$ lacks line, column, and block IDs.	Two-column fax row: `FAX: (336)...  | PHONE: ...`	**Hierarchical Position Encoding (Line, Block, Column IDs)**
2. Mid-Span Noise	Multi-word entities fracture mid-line into noisy BIO tags.	Multi-word sender field: `June Flynn for Eric Brown/...`	**Segment Context Module (Mean Pool + ReZero Gate)**
3. Segment Drift	Pure segment pooling overwrites subword boundaries.	Receipt total line: `TOTAL (Qty 2.00) 60.000`	**Masked Gated Late-Fusion (Anchor Lock + Subword Gate)**
4. Span Merging	Adjacent key–value entities on one line merge together.	Form row on one line: `TO: George Baroody`	**Intra-Line Boundary Loss (Adjacent Token Pair BCE)**

2. Hierarchical Position Encoding (HPE) – Visual Grouping



1. Line Grouping (`Line 1..5`)

Boxes with $> 50\%$ vertical overlap share the same horizontal reading line ID.

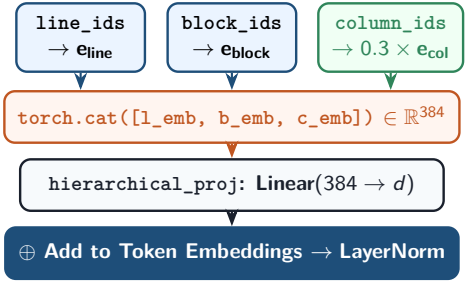
2. Column Detection (`Column 0..1`)

Horizontal jumps $> 12\%$ of page width separate multi-column form fields.

3. Block Grouping (`Block 0..1`)

Consecutive lines with vertical gap $< 1.8\times$ line height form a semantic block.

2. Hierarchical Position Encoding (HPE) – Core Implementation



LayoutLMv3Embeddings.forward (modeling_layoutlmv3.py):

```
def forward(
    self, input_ids: Tensor, bbox: Tensor,
    line_ids: Optional[Tensor],
    block_ids: Optional[Tensor],
    column_ids: Optional[Tensor], ...
) -> Tensor:
    ...
    l_emb = self.line_position_embeddings(line_ids)
    b_emb = self.block_position_embeddings(block_ids)
    c_emb = self.column_position_embeddings(column_ids) * 0.3
    concat_emb = torch.cat([l_emb, b_emb, c_emb], dim=-1)
    embeddings = embeddings + self.hierarchical_proj(
        concat_emb)
    return self.dropout(self.LayerNorm(embeddings))
```

3. Segment Context Module (SCM) – Visual Concept



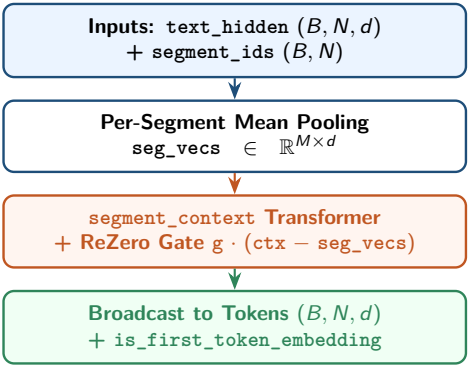
1. Mean Pool
Average all 8 tokens in segment $\rightarrow s_k$

2. Segment Attention
1-Layer 4-Head Transformer $\rightarrow c_k$

3. ReZero Gate (g)
Smooth residual blend
$s_k + g(c_k - s_k)$

4. Broadcast + Onset
Tag June as `B-ANS`
& rest as `I-ANS`

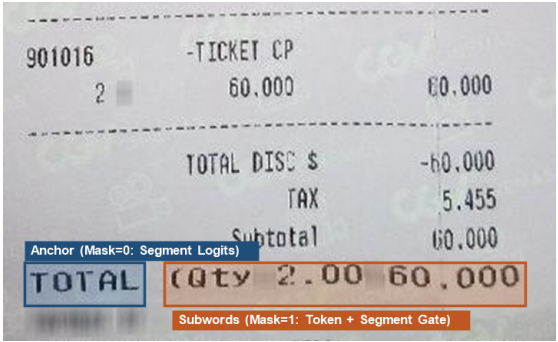
3. Segment Context Module (SCM) – Core Implementation



_segment_pool_and_contextualize (modeling_layoutlmv3_segment.py):

```python
def _segment_pool_and_contextualize(
    self, text_hidden: Tensor, segment_ids: Tensor
) -> Tensor:
    ...
    seg_vecs = torch.stack([
        text_hidden[b, seg_b == sid].mean(dim=0)
        for sid in unique_segs], dim=0)
    ctx = self.segment_context(
        seg_vecs.unsqueeze(0)).squeeze(0)
    seg_vecs = seg_vecs + self.segment_context_gate * (
        ctx - seg_vecs)
    ...
    is_first[:, 1:] = (
        segment_ids[:, 1:] != segment_ids[:, :-1]).long()
    return seg_repr + self.is_first_token_embedding(is_first)
```

4. Masked Gated Late-Fusion – Visual Concept



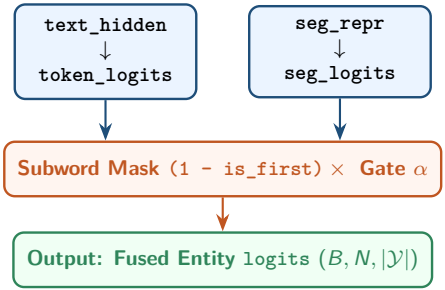
Case 1: Segment Anchor (`Mask = 0`)

First token of segment (`TOTAL`):
100% Segment Logits $\rightarrow$ Locks anchor class to segment consensus.

Case 2: Internal Subwords (`Mask = 1`)

Continuation tokens (`(Qty 2.00) 60.000`):
Gated Blend (α) of Segment Logits + Token Shortcut Logits for fine subword precision.

4. Masked Gated Late-Fusion – Core Implementation

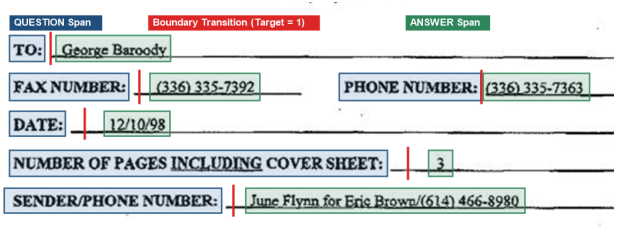


LayoutLMv3ForSegmentTokenClassification.forward:

```
def forward(
    self, input_ids: Tensor, bbox: Tensor,
    segment_ids: Tensor, ...
) -> TokenClassifierOutput:
    ...
    token_logits = self.token_classifier(
        self.dropout(text_hidden))
    seg_logits = self.classifier(self.dropout(seg_repr))

    subword_mask = (1 - is_first).unsqueeze(-1).float()
    alpha = self.alpha_max * torch.sigmoid(self.fusion_gate)
    logits = seg_logits + subword_mask * alpha * (
        token_logits - seg_logits)
    ...
```

5. Intra-Line Boundary Loss – Visual Concept



Adjacent Pair Inside Same Span

Example: `["FAX", "NUMBER:"]`

Same Entity ID $\Longrightarrow$ **Target = 0**

Adjacent Pair Across Boundary

Example: `["NUMBER:", "(336)"]`

Entity Changes (`QUESTION` $\rightarrow$ `ANSWER`)

$\Longrightarrow$ **Target = 1** (Sharpens Span Split)

5. Intra-Line Boundary Loss – Core Implementation

Adjacent Token States

$\mathbf{h}_i$ = `text_hidden[:, :-1]`

$\mathbf{h}_{i+1}$ = `text_hidden[:, 1:]`

↓

Concatenate $[\mathbf{h}_i \ ; \ \mathbf{h}_{i+1}] \in \mathbb{R}^{2d}$

→ **2-Layer MLP** `boundary_classifier`

↓

Compare (`ent_i` $\neq$ `ent_next`)

→ **Binary Cross-Entropy Loss**

`_compute_boundary_loss`

(`modeling_layoutlmv3_segment.py`):

```python
def _compute_boundary_loss(
    self, text_hidden: Tensor,
    entity_ids: Tensor, labels: Tensor
) -> Tensor:
    h_pair = torch.cat([
        text_hidden[:, :-1, :],
        text_hidden[:, 1:, :]], dim=-1)
    b_logits = self.boundary_classifier(h_pair).squeeze(-1)
    ...
    b_labels = (
        entity_ids[:, :-1] != entity_ids[:, 1:]).float()
    return F.binary_cross_entropy_with_logits(
        b_logits[valid], b_labels[valid])
```

6. Experimental Setup: Hardware, Environment & Datasets

Training Environment

GPU Server	**NVIDIA A100-SXM4-80GB**
System RAM	167 GiB Host Memory
Environment	Ubuntu Linux, Python 3.7.16
DL Stack	PyTorch 1.10.0+cu111, FP16
Backbone Lib	Transformers 4.12.5
Metric Lib	`seqeval` 1.2.2 (Strict BIO)
Random Seeds	3 Seeds: $\{42, 123, 1993\}$

Benchmark Datasets

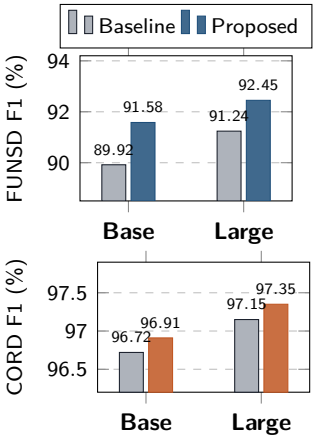
Attribute	**FUNSD**	**CORD**
Document Type	Scanned Forms	Retail Receipts
Total Images	199	1,000
Train / Val	149 / –	800 / 100
Test Split	50 pages	100 pages
Entity Classes	3 (7 BIO tags)	30 (61 BIO tags)
Layout Style	Multi-column	Tabular rows

6. Run Configurations & Hyperparameters

Configuration Parameter	FUNSD (base)	CORD (base)	FUNSD (large)	CORD (large)
Pretrained Checkpoint	`microsoft/layoutlmv3-base`		`microsoft/layoutlmv3-large`	
Architecture Scale	12 Layers, $d = 768$ (125M)		24 Layers, $d = 1024$ (356M)	
Optimizer & Weight Decay	AdamW ($\beta_1 = 0.9, \beta_2 = 0.999$, Weight Decay $= 0.01$)			
Training Steps & Warmup	$1,000$ Steps (10% Linear Warmup $= 100$ Steps)			
Backbone Learning Rate	5×10^{-5}	5×10^{-5}	1×10^{-5}	5×10^{-5}
Specialized Module LR	5×10^{-4} (HPE, SCM, Late-Fusion, Boundary Classifier)			
Effective Batch Size	16	64	16	16
HPE Max ($N_{\text{line}}, N_{\text{block}}, N_{\text{col}}$)	$(50, 20, 10)$	$(80, 15, 8)$	$(50, 20, 10)$	$(80, 15, 8)$

7. Experimental Results: Baseline vs. Proposed Model (3 Seeds)

Model	Data	Prec. (%)	Rec. (%)	F1 (%)
Baseline (base)	FUNSD	89.32 ± 1.31	90.53 ± 0.48	89.92 ± 0.85
Proposed (base)	**FUNSD**	$\mathbf{91.82 \pm 0.05}$	$\mathbf{91.33 \pm 0.28}$	$\mathbf{91.58 \pm 0.16}$
Baseline (large)	FUNSD	90.69 ± 1.01	91.80 ± 0.23	91.24 ± 0.62
Proposed (large)	**FUNSD**	$\mathbf{92.31 \pm 0.51}$	$\mathbf{92.60 \pm 0.10}$	$\mathbf{92.45 \pm 0.26}$
Baseline (base)	CORD	96.50 ± 0.39	96.94 ± 0.28	96.72 ± 0.33
Proposed (base)	**CORD**	$\mathbf{96.80 \pm 0.27}$	$\mathbf{97.02 \pm 0.15}$	$\mathbf{96.91 \pm 0.21}$
Baseline (large)	CORD	96.96 ± 0.30	97.35 ± 0.04	97.15 ± 0.17
Proposed (large)	**CORD**	$\mathbf{97.28 \pm 0.04}$	$\mathbf{97.43 \pm 0.04}$	$\mathbf{97.35 \pm 0.04}$



7. Component Ablation Study (3-Seed Mean ± Std F1 %)

Ablation Configuration	FUNSD (base)	CORD (base)	FUNSD (large)	CORD (large)
1. Baseline (Vanilla LayoutLMv3)	89.92 ± 0.85	96.72 ± 0.33	91.24 ± 0.62	97.15 ± 0.17
2. Full Model (HPE + SCM + Bound)	$\mathbf{91.58 \pm 0.16}$	$\mathbf{96.91 \pm 0.21}$	$\mathbf{92.45 \pm 0.26}$	$\mathbf{97.35 \pm 0.04}$
3. Ablation: w/o Boundary Loss	91.62 ± 0.19	97.06 ± 0.18	92.15 ± 0.24	97.01 ± 0.42
4. Ablation: w/o HPE	91.33 ± 0.27	97.10 ± 0.15	92.69 ± 0.18	–
5. Ablation: w/o Segment (SCM)	$\mathbf{90.36 \pm 0.30}$	$\mathbf{96.59 \pm 0.57}$	$\mathbf{91.01 \pm 0.23}$	$\mathbf{96.84 \pm 0.13}$

Impact of Removing SCM

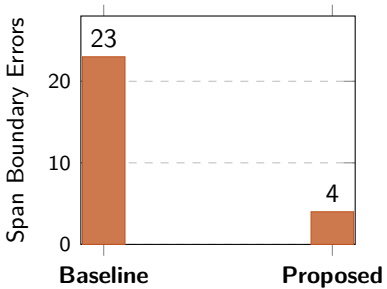
Steepest F1 drop across all settings: $-$**1.44%** on FUNSD-large and $-$**1.22%** on FUNSD-base.

Impact of Removing Boundary Loss

Degrades large F1 by $-$**0.30%** (FUNSD) and $-$**0.34%** (CORD), increasing CORD std 10×.

7. Entity-Level Error Breakdown on FUNSD (1,998 Test Entities)

Entity Outcome Category	**Baseline**	**Proposed**	**Diff.**
Exact Span & Class (TP ↑)	1,806	**1,830**	**+24**
Span Boundary Error (↓)	23 (19/4)	**4** (4/0)	**−82.6%**
Missed Entities (FN ↓)	79	**71**	**−8**
Spurious Predictions (FP ↓)	72	**59**	**−13**



8. Future plans

1. **Fine-Tuning on FUNSD and CORD Datasets**
2. **Final Ablation Verification and Visual Case Studies**
3. **Continuous In-Depth Review of Thesis Papers**



Thank You for Your Attention!

Special thanks to Supervisor

PhD. Pham Thi Quynh Trang

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